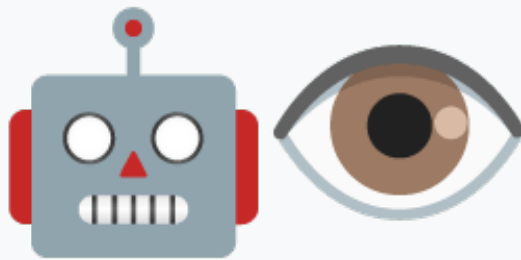


Planrs Academy Master Series

EYES OF THE MACHINE

How AI Sees and Understands the World



Grade 6 | Session 1

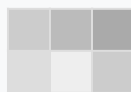
The AI Eye Test

Can you spot what an object is before the AI does? [cite: 88]

Look at a blurry image of a famous landmark—like the **Taj Mahal** or **India Gate**. As the image becomes clearer, the AI starts guessing. [cite: 89]

Machines don't see things the same way we do. While we see a beautiful building, the AI sees millions of tiny dots! [cite: 92]

Example of a "blurry" pixelated view:



What is Computer Vision?

Computer Vision is how machines learn to **'see'** and **'understand'** pictures—just like your eyes! [cite: 90, 91]

AI looks at images and finds patterns by breaking pictures into tiny dots called **pixels**. [cite: 92]

How it works:

- The camera takes a picture.
- The AI turns the picture into numbers based on colors.
- The AI searches for shapes like circles (for wheels) or triangles (for a Samosa!).

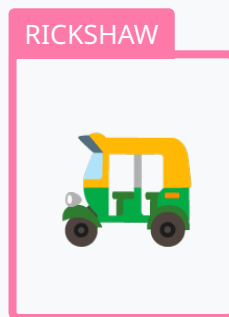
Bounding Boxes: The AI's Highlighter

Bounding Boxes are colored rectangles that AI draws around objects it finds in images. [cite: 95, 96]

Just like you use a highlighter to mark important words, AI uses bright boxes to show "**I found something here!**" [cite: 97]

Each box surrounds one specific object, such as: [cite: 98]

- An Auto-Rickshaw [cite: 98]
- A Bicycle [cite: 98]
- Even a Cow on the street! [cite: 98]



Classification vs. Detection

There is a key difference between just naming something and finding exactly where it is.
[cite: 113]

1. Classification: "What is it?" [cite: 108, 109]

AI identifies the main object in the image. *Example: "This is a Mango."* [cite: 110]

2. Detection: "Where is it?" [cite: 111, 112]

AI finds **AND** locates every instance. *Example: "There are 5 Mangoes here, and here is each location."* [cite: 112, 113]

Confidence Scores

A Confidence Score tells us how sure the AI is about what it sees. [cite: 115, 116]

Think of it like a test score—a higher percentage means the AI is more certain about its answer! [cite: 117]

Samosa Detected: 95% Confidence [cite: 118]



The AI is very sure! [cite: 118]

Kachori Detected: 60% Confidence [cite: 119]



The AI is less certain. [cite: 119]

Interactive Demo: Indian Kitchen

Let's look at a typical Indian kitchen counter to see what the machine's eyes can find.
[cite: 131, 132]

Notice the neon bounding boxes around each object with labels and confidence scores!
[cite: 133]

- **Plate:** 91% Confidence [cite: 135]
- **Tumbler:** 91% Confidence [cite: 136]
- **Samosas:** Detected with various scores! [cite: 118, 122]

Try to draw your own boxes around a glass, a spoon, or a rolling pin (belan) in your mind!

How AI Helps on Indian Roads

AI detects Rickshaws, Cycles, and Cows to keep roads safe! [cite: 137, 138]

Traffic cameras and systems like **FASTag** use computer vision to spot different vehicles and animals on busy streets. [cite: 139]

This helps: [cite: 140]

- Manage traffic flow. [cite: 140]
- Prevent accidents. [cite: 140]
- Detect vehicle numbers automatically. [cite: 139]

AI in Indian Agriculture: Smart Farming

AI helps farmers by detecting **ripe** and **unripe** Alphonso mangoes using computer vision and bounding boxes. [cite: 151, 153]

This is important because it helps farmers: [cite: 154]

- Pick the right fruit at the right time. [cite: 154]
- Reduce waste. [cite: 154]
- Increase profits by selling top-quality fruit! [cite: 154]



Counting Crowds at Festivals

AI counts people at **Melas** and festivals to help manage crowds safely. [cite: 158, 159]

Bounding boxes around groups help organizers know how many people are in each area. [cite: 160]

This technology ensures that everyone has a safe and happy time during large celebrations like Diwali or Ganesh Chaturthi! [cite: 159, 160]

Can You Spot the Error?

Look carefully! Sometimes the AI draws a bounding box where there is no object, or it gets the name wrong. [cite: 162, 163]

Why do you think AI might get confused sometimes? [cite: 165]

- Bad lighting (too dark or too bright).
- Objects overlapping each other.
- Seeing a shadow that looks like a person!

AI is smart, but it still needs humans to teach it! [cite: 164]

Recap: What We Learned Today

- **Computer Vision:** Machines "seeing" like us. [cite: 167]
- **Bounding Boxes:** Colorful rectangles showing WHERE objects are. [cite: 168]
- **Confidence Scores:** Numbers telling HOW SURE the AI is. [cite: 169]

Your Turn: Imagine AI's Eyes

Where else in India could AI's eyes help people? [cite: 170]

Think about your school, your neighborhood, or places you visit. What could computer vision detect and help with? [cite: 171]